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Progressive Disclosure: the technique that helps control context (and tokens) in AI agents

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If you closely follow the evolution of AI agents, you have probably heard about the concept of *Agent Skills*. One of the most interesting architectural ideas behind this approach is the use of **Progressive Disclosure** applied to the agent's internal operation itself.

Although it now appears in LLM-based systems, the concept is not new. It originates from user interface design, where it was used to show only the necessary information at each moment, hiding complexity until the user actually needed it. Decades later, the same principle is being reused to manage one of the most critical resources in modern AI: **context**.

What does Progressive Disclosure mean in AI agents?

Instead of loading all available capabilities into the model's context from the beginning, agents organize their tools or skills into different levels of detail.

The agent does not automatically access all information. It first evaluates which capabilities might be relevant and only then incorporates additional data into the context.

In practice, this is usually structured in layers:

Layer 1 — Discovery

The agent only sees lightweight metadata: the skill name and a short description. This allows it to know which capabilities exist without unnecessarily increasing context size.

Layer 2 — Activation

If a skill appears useful for the task, the system loads the instructions required to use it correctly.

Layer 3 — Execution / Reference

Only when necessary are examples, extensive documentation, or additional reference information added.

This process typically does not rely solely on the model; additional mechanisms such as routing, embedding-based retrieval, or planners usually help decide which information should be loaded.

Why can it reduce token usage?

Language models process all the context they receive. If an agent always included the full definition of dozens of tools, costs would quickly grow and the context would become saturated.

With Progressive Disclosure:

- the agent first loads minimal information,
- expands context only when a capability is relevant,
- avoids introducing unnecessary instructions.

The savings depend on how many skills exist and how many are actually used per task, but they can be significant in complex systems.

It is important to clarify that the main goal is not simply saving tokens; cost reduction is a consequence of better context management.

Clean context ≠ less information

A common — and somewhat simplified — idea is that "giving the model less information produces better results." A more accurate statement would be:

Models perform better when they receive the relevant information at the right moment.

An overloaded context containing irrelevant data can cause:

- higher computational cost,
- increased latency,
- distractions during reasoning,
- poorer tool selection.

However, removing useful information also degrades performance. The real challenge is not minimizing context, but optimizing its composition.

Beyond savings: why this pattern matters

Progressive Disclosure also improves several key aspects of agent design:

- reduces decision complexity during tool selection,
- limits the number of simultaneous instructions the model must consider,
- enables modular and scalable architectures,
- allows new skills to be added without constantly inflating the base context.

For this reason, it increasingly appears alongside techniques such as:

- tool routing,

- hierarchical planning (planner–executor),
- selective information retrieval,
- dynamic context compression.

Is it already a standard?

Not completely. Although the pattern is rapidly gaining adoption, there is still no universal implementation. Different frameworks and architectures apply their own variations of the same principle.

What does seem clear is that as agents incorporate more capabilities, carefully managing what information enters the context stops being an optional optimization and becomes an architectural necessity.

Summary

An LLM's context is a limited resource. Progressive Disclosure is not about providing less information, but about revealing it progressively according to the agent's actual needs.

When applied correctly, it enables systems that are more efficient, more scalable, and often cheaper to operate.

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